TRADEWINDS RESUBMISSION DOSSIER // DOCKET 10-26-3976 (RESUBMISSION OF 9-26-3703)

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DOD CDAO TRADEWINDS SOLUTIONS MARKETPLACE // RESUBMISSION DOSSIER & VIDEO SCRIPT
ACTIVE DOCKET: 10-26-3976 (RESUBMISSION OF 9-26-3703) | READINESS: TRL 4 (TARGET: TRL 6)
CONTRACTOR: Humphrey Virtual Farms LLC | CAGE: 1AHA8 | UEI: S1M4ENLHTDH5

PART 1: 4-PILLAR RESOLUTION OF PRIOR DOCKET 9-26-3703 ASSESSOR FEEDBACK

[CRITERION-1] CRITIQUE : Defining the Problem: Quantify exact DoD infrastructure exposure, federal threat citations, and physical scan-latency failure.
RESOLUTION: Cited U.S. GAO reports GAO-21-81 & GAO-24-106137 (500+ DoD installations, 300,000+ buildings, $4.2B+ OT energy assets on 50-200ms PLCs) and Joint CISA/NSA/DoD Advisories AA23-144A & AA24-038A (Volt Typhoon OT pre-positioning) plus TRISIS/Triton. Proved physics gap: <4.16ms arc-flash backfeeds and 6.0ms microreactor excursions outrun 50-200ms PLCs by 8x to 33x.

[CRITERION-2] CRITIQUE : Accelerating the Mission: Clarify TRL 4 software testbed vs. TRL 6/7 hardware target, zero cloud dependency, and empirical test metrics.
RESOLUTION: Explicitly baselined at TRL 4 (Laboratory Software Emulation Testbed, Phase-1 OTA Target: TRL 6) with 0% cloud dependency on 127.0.0.1 loopback (100% EMCON availability). Brain 1: <50us loop, 2.04us Modbus FC05, 13.33ms breaker isolation, 126.13ms soft-start. Brain 2: <25W NPU (<5.0ms, advisory only, 0 actuator wire). Brain 3: Ed25519/SHA-256 WORM DAG (<0.15ms, 52 tables). Verified by 636 C2 modules (1,961 Private / 1,770 Public), 85/85 passing test suites, 17/17 Red-Team blocks, and 1,420 Academy Qs.

[CRITERION-3] CRITIQUE : Advancing the State of the Art: Provide a quantitative head-to-head Analysis of Alternatives (AoA) against commercial OT/SCADA & AI incumbents.
RESOLUTION: Benchmarked Project Ebony’s modeled 36.75ns Analog Kinetic Guillotine MOSFET cutoff (1,360,544x faster than a 50ms PLC, 28.5W on 24V DC, 0% cloud) directly against Siemens S7-1500/Rockwell ControlLogix (50-200ms, 150-450W), Schneider Triconex SIS (10-50ms, 200-600W), passive IDS Dragos/Nozomi Guardian (0 physical cutoff authority, 300-1,200W), and Jev Cloud AI (>250ms RTT).

[CRITERION-4] CRITIQUE : Business Model & Validation: Document concrete commercial/government traction, unbundled per-node CLIN pricing, and 10 U.S.C. 4022 OTA transition.
RESOLUTION: Documented Target Commercialization Roadmap validation (30% water savings, 12% yield gain), State of Oklahoma executive briefings (OK Commerce CTO Rob Teel & CEO John Budd under HB 2992 & Title 61, node HVF-NODE-TEELE-001, OMES Title 74/M254 filings, $25K ODOT bridge drone cutting $4,600 inspections to $250/sortie with $500K allowable in-kind match under 2 CFR 200.306), 26-Module CLIN Schedule (CLIN 0001-0026;

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$895K/yr 1-Node Bundle; $2.45M/yr 25-Node Base Fleet), and $1.485M 10 U.S.C. 4022 OTA -> 4022(f) production.

PART 2: QUANTITATIVE ANALYSIS OF ALTERNATIVES (COMPETITOR COMPARISON MATRIX)

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* METRIC: Primary Architecture
- Project Ebony (HVF 1AHA8) : Tri-Brain Bare-Metal + 36.75ns Modeled Analog Guillotine (TRL 4 Testbed)
- Siemens / Rockwell PLCs : Digital OS/RTOS PLC (5-20Hz Cyclic Scan)
- Dragos / Nozomi Guardian : Passive Network IDS / Cloud-Hybrid SIEM
- Schneider Triconex SIS : Triple-Modular Redundant Digital PLC
- Cloud AI Wrappers (Jev) : Cloud API Classification Wrapper
* METRIC: Physical Cutoff Latency
- Project Ebony (HVF 1AHA8) : 36.75 ns (Modeled Analog) / 2.04 us (Modbus FC05) / 13.33 ms Breaker
- Siemens / Rockwell PLCs : 50,000,000 - 200,000,000 ns (50 - 200 ms)
- Dragos / Nozomi Guardian : NONE (0 Physical Cutoff Authority)
- Schneider Triconex SIS : 10,000,000 - 50,000,000 ns (10 - 50 ms)
- Cloud AI Wrappers (Jev) : 250,000,000+ ns (> 250 ms Cloud API RTT)
* METRIC: Speed & Physics Advantage
- Project Ebony (HVF 1AHA8) : 1,360,544x Faster than 50ms PLC (Arrests <4.16ms Arc-Flash)
- Siemens / Rockwell PLCs : 8x - 33x Too Slow for <4.16ms Arc-Flash / 6ms Excursion
- Dragos / Nozomi Guardian : Requires Human Operator (> 250 ms Reaction)
- Schneider Triconex SIS : Vulnerable to TRISIS/Triton Firmware Injection
- Cloud AI Wrappers (Jev) : Total Failure Under EMCON / SATCOM Severance
* METRIC: Total SWaP Power Draw
- Project Ebony (HVF 1AHA8) : 28.5 W (< 30W on 24V DC Tactical Battery Bus)
- Siemens / Rockwell PLCs : 150 W - 450 W (Rack + HMI + Active Cooling)
- Dragos / Nozomi Guardian : 300 W - 1,200 W (Appliance Server + Cloud)
- Schneider Triconex SIS : 200 W - 600 W (TMR Chassis)
- Cloud AI Wrappers (Jev) : Data Center GPU Cluster Required
* METRIC: Cloud / OS Dependency
- Project Ebony (HVF 1AHA8) : 0% (100% Air-Gapped Localhost 127.0.0.1 / RS-485 Loopback)
- Siemens / Rockwell PLCs : Proprietary Firmware / Windows Engineering Workstation
- Dragos / Nozomi Guardian : High (Cloud Threat Feeds / Linux OS)
- Schneider Triconex SIS : Proprietary Firmware (Compromised in TRISIS)
- Cloud AI Wrappers (Jev) : 100% Dependent on External Commercial Cloud
* METRIC: WORM Cryptographic Ledger

- Project Ebony (HVF 1AHA8) : Ed25519 + SHA-256 WORM Merkle DAG (<0.15ms | 52 SQLite Tables)
- Siemens / Rockwell PLCs : Unauthenticated Modbus/CIP by Default
- Dragos / Nozomi Guardian : Mutable Syslog / SIEM Export
- Schneider Triconex SIS : Standard Event Buffer (Non-Merkle)
- Cloud AI Wrappers (Jev) : Mutable Cloud Database Logs

PART 3: SECTION VII.e COMPLIANT VIDEO TELEPROMPTER SCRIPT (5 SEGMENTS)

>>> SLIDE 1 (0:00 - 0:50) // RESUBMISSION GATE: 4-PILLAR ASSESSOR FEEDBACK RESOLUTION
RUBRIC: Mandatory Resubmission Compliance Gate (Docket 10-26-3976 Resubmission of 9-26-3703)
SCRIPT: I am Jeffery Humphrey, Founder, President, and CEO of Humphrey Virtual Farms LLC, CAGE Code 1AHA8, UEI S1M4ENLHTDH5. This video is our official resubmission for Project Ebony under Tradewinds Docket 10-26-3976, resolving all four pillars of Assessor Feedback from our prior submission 9-26-3703. First, we explicitly baseline Project Ebony at TRL 4--a laboratory software emulation testbed and mathematical proof of concept running on air-gapped loopback 127.0.0.1, targeting TRL 6 operational field demonstration under our proposed 12-month, $1.485 million Phase-1 Prototype OTA. Second, we quantify the exact DoD threat landscape under GAO-21-81, GAO-24-106137, and CISA Volt Typhoon advisories. Third, we provide a side-by-side Quantitative Analysis of Alternatives against Siemens, Rockwell, Schneider Triconex, Dragos, Nozomi, and cloud AI wrappers. Fourth, we detail our State of Oklahoma executive validation and our 26-module unbundled CLIN price schedule.

>>> SLIDE 2 (0:50 - 1:40) // CRITERION 1: DEFINING THE DOD PROBLEM (GAO & CISA VOLT TYPHOON)
RUBRIC: Criterion 1: Defining the Problem (Upgraded from Acceptable to Outstanding)
SCRIPT: Criterion 1: Defining the Problem. Per U.S. GAO reports GAO-21-81 and GAO-24-106137, the Department of Defense operates over 500 military installations, 300,000 buildings, and $4.2 billion in annual installation energy and microgrid assets relying on legacy commercial SCADA PLCs scanning at 50 to 200 milliseconds. Per Joint CISA, NSA, FBI, and DoD Advisories AA23-144A and AA24-038A, state-sponsored Volt Typhoon actors actively pre-position across critical energy, water, and transportation OT corridors. In TRISIS and Industroyer2, attackers corrupted controller logic while spoofing operator HMIs. Because high-voltage arc-flash backfeeds propagate in under 4.16 milliseconds and microreactor excursions occur in 6.0 milliseconds, 50-to-200 millisecond commercial PLC scan loops are 8 to 33 times too slow to arrest kinetic equipment destruction.

>>> SLIDE 3 (1:40 - 2:40) // CRITERION 2: ACCELERATING THE MISSION (TRI-BRAIN TRL 4 TESTBED)
RUBRIC: Criterion 2: Accelerating the Mission (Zero-Cloud Tri-Brain & 85/85 Test Suite)
SCRIPT: Criterion 2: Accelerating the Mission. Project Ebony version 8.0.0 operates with zero percent commercial cloud dependency, zero external APIs, and zero WAN listening ports on air-gapped 127.0.0.1 loopback, maintaining 100 percent availability under EMCON or SATCOM severance. Brain One executes a deterministic control loop in under 50 microseconds, achieving 2.04-microsecond RS-485 Modbus FC05 actuation, 13.33-millisecond 4-channel breaker isolation, and 126.13-millisecond sequential soft-start recovery. Brain Two runs sub-25-watt local NPU harmonic anomaly detection in under 5 milliseconds, wired as strictly advisory with zero electrical connection to actuator coils. Brain Three seals every frame in under 0.15 milliseconds into an Ed25519 and SHA-256 WORM Merkle DAG across 52 bare-metal SQLite tables, verified by 636 C2 modules, 85 of 85 passing test suites, 17 of 17 blocked Red-Team cyber vectors, and 1,420 verified Field Academy exam questions.

>>> SLIDE 4 (2:40 - 3:40) // CRITERION 3: ADVANCING STATE OF THE ART (COMPETITOR AoA **MATRIX)**
RUBRIC: Criterion 3: Quantitative Analysis of Alternatives vs. Legacy PLCs, IDS & Cloud AI
SCRIPT: Criterion 3: Advancing the State of the Art. On screen is our Quantitative Analysis of Alternatives comparing Project Ebony directly against incumbent competitors. Siemens S7-1500 and Rockwell ControlLogix PLCs require 50 to 200 milliseconds per digital scan and draw 150 to 450 watts. Schneider Electric Triconex safety systems require 10 to 50 milliseconds, draw 200 to 600 watts, and were breached by firmware injection in TRISIS. Passive intrusion detection systems like Dragos Platform and Nozomi Guardian draw 300 to 1,200 watts and have zero physical cutoff authority, requiring over 250 milliseconds for human intervention, just like cloud AI wrappers such as Jev that fail when air-gapped. By contrast, Project Ebony’s modeled Analog Kinetic Guillotine sweeps MOSFET gate charge in 36.75 nanoseconds--1,360,544 times faster than a 50-millisecond PLC--at just 28.5 watts on a 24-volt DC battery bus with cryptographic WORM provenance.

>>> SLIDE 5 (3:40 - 4:35) // CRITERION 4: BUSINESS MODEL, OKLAHOMA VALIDATION & 26-CLIN SCHEDULE
RUBRIC: Criterion 4: Customer Validation, 26-Module CLIN Pricing & $1.485M 10 U.S.C. **4022 OTA**
SCRIPT: Criterion 4: Business Model and Customer Validation. Project Ebony is validated across our commercial agricultural and microgrid roadmap targeting 30 percent water savings and 12 percent yield gains, alongside October 2026 executive briefings with the Oklahoma Department of Commerce CTO Rob Teel and CEO John Budd under Oklahoma HB 2992

and Title 61, node HVF-NODE-TEELE-001, OMES supplier filings, and a $25,000 Phase-1 ODOT bridge inspection drone prototype slashing $4,600 snooper-truck inspections to $250 per sortie across 8 ODOT divisions with a $500,000 allowable in-kind match under 2 CFR 200.306. Our unbundled 26-module console scales from CLIN 0001 Base Pilot at $2,450 per month and CLIN 1004-A Full 1-Node Bundle at $895,000 per year to our $1.485 million, 12-month Fixed-Price Prototype OTA under 10 U.S.C. Section 4022, transitioning directly to non-competitive production under Section 4022(f) while retaining 100 percent DFARS 252.227-7018 Background IP. Thank you.